

SMP Upstream Integration Plan

Step-by-step migration to upstream xpack-development

Multi-Core SMP Upstream Integration Plan: Exhaustive Step-by-Step Technical Reference

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Multi-Core SMP Upstream Integration Plan: Exhaustive Step-by-Step Technical Reference

A Granular, Non-Breaking Strategy for Integrating `smp` into

`xpack-development`

Status (2026-10-01) — this is the strategy; the execution lives elsewhere. This document is the *reference plan* (why, and the per-step technical detail). The **concrete, executable, and now dry-run-verified** companion is `Implementation-SMP-Integration.md` (`.pdf`): real scripts in `scripts/smp/`, byte-reproducible per-step recipes, the acceptance gate, and (new §11) the plain-English playbook plus the repeatability contract. The full 30-step series has been dry-run end to end **locally, with nothing pushed**.

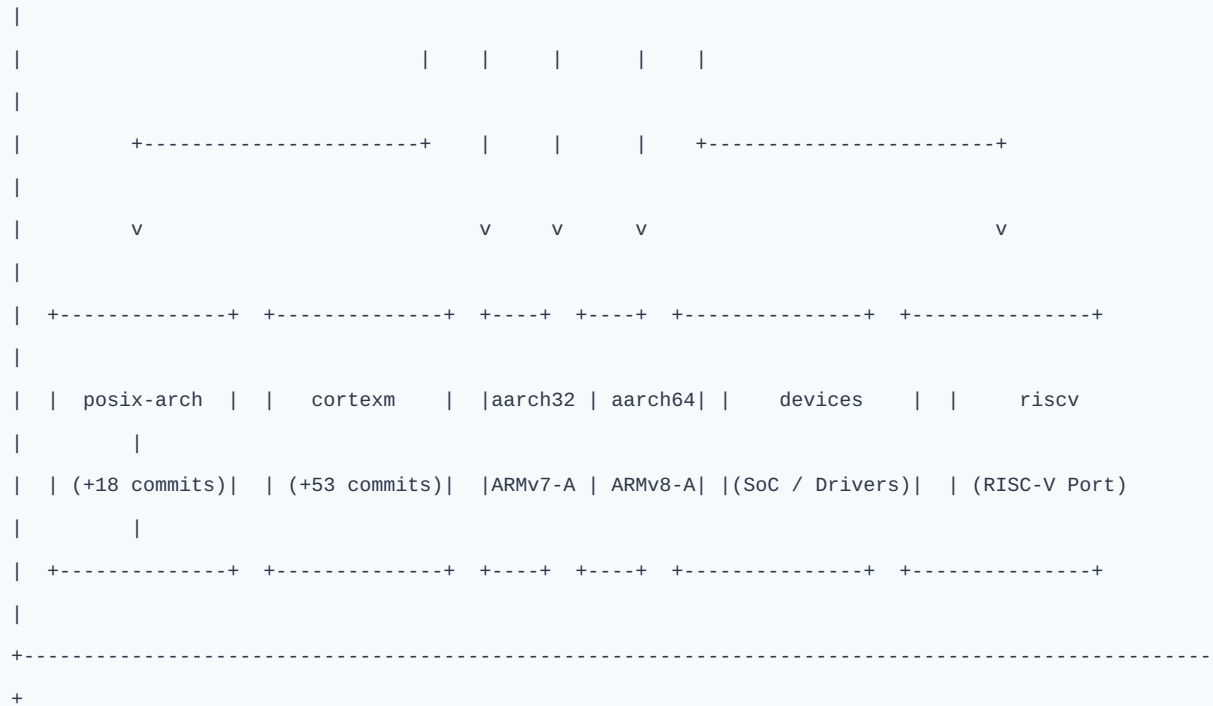
A few places where the executed reality refined this plan — prefer the Implementation doc where they differ: - **devices is dissolved first (Part 0)** into the architecture ports; the integrated cortexm/posix-arch CMake links the fat `micro-os-plus::iii` and pull **no** `micro-os-plus-iii-devices` sibling (the `smp` branch's standalone `UOS_SMP_DIR / add_subdirectory` model is *not* used). - **The verification harness is** `scripts/smp/verify-step.sh` (with `check-pristine.sh`, `sc-project.py`, `link-ports.sh`, ...), not the inline sketch reproduced in §4 below; the real one is part-aware (frozen in A/B, additive-only in C) and FAST-mode capable. - **SMP must be enabled via a platform's** `target_compile_definitions`, never a `cmake -D` cache variable (the latter silently compiles single-core). - **Nothing self-publishes:** `release-port.sh` bumps versions in-place (avoiding `npm version`'s push hook) and `finalize.sh` stops before any push.

1. Architectural Baseline & Repository Topology

1.1 The Ecosystem Landscape

The μ OS++ IIIe Real-Time Operating System is structured as a modular, multi-repository architecture where a portable core library communicates with target-specific hardware ports:





1.2 Divergence Profile & Summary

Repository / Component	Upstream Baseline (<code>xpack-development</code>)	Multi-Core Branch (<code>smp</code>)	Delta / Characteristics
<code>micro-os-plus-iii</code>	Commit <code>3a22a06b</code> (2026-09-20)	Commit <code>1a728371</code> (2026-09-29)	+143 commits, 43 modified files in <code>include/</code> and <code>src/</code> . Zero new kernel source files added.
<code>micro-os-plus-iii-posix-arch</code>	Tag <code>v1.0.1</code>	<code>smp</code> (+18 commits)	Host-thread-as-CPU model, per-CPU signal masking, compiler barriers.
<code>micro-os-plus-iii-cortexm</code>	Tag <code>v1.1.0</code>	<code>smp</code> (+53 commits)	SysTick ICSR fix, ARMv8-M / Cortex-M33, RP2350 SIO hardware spinlocks.
<code>micro-os-plus-iii-aarch32</code>	Upstream tracking	<code>smp</code> active	RPi3B and RPi Zero 2W support, GICv2 distributor & CPU interface.
<code>micro-os-plus-iii-aarch64</code>	Upstream tracking	<code>smp</code> active	64-bit ARMv8-A execution, GICv2/v3 support, AArch64 MMU translation.
<code>micro-os-plus-iii-devices</code>	Does not exist upstream	<code>smp</code> active	Shared device abstractions, board console mirror, clock definitions.
<code>micro-os-plus-iii-riscv</code>	Upstream tracking	<code>smp</code> active	RISC-V CLINT / PLIC machine timer and software interrupt handling.

2. Inviolable Migration Rules & Upstream Constraints

2.1 The Core Migration Rules

1. **Granular Code Chunks, Not File Copies:** A step consists of small, logically cohesive code snippets (a single function fix, struct definition, or conditional block).
2. **Cross-Repository Synchronization:** When a kernel change requires port support, the kernel and the corresponding ports (`posix-arch`, `cortexm`) are updated and linked in the exact same step.
3. **Logical Progression:**
4. **Part A (Steps 1–13):** Uniprocessor bug fixes, memory safety, C++17 conformance, and POSIX I/O hardening (compiled and executed by existing tests).
5. **Part B (Steps 14–23):** Multi-core SMP infrastructure, strictly gated behind `#if defined(OS_USE_SMP_SCHEDULER)` (zero single-core delta verified via `unifdef`).
6. **Part C (Steps 24–28):** Port releases, new architecture files, CMake targets, and add-only test harness extensions.
7. **Part D (Steps 29–30):** Documentation, Typst PDFs, and final merge.
8. **Pristine Test Procedures:** Existing test suites, platforms, and `package.json` action workflows in `xpack-development` **must never be modified**.
9. **Continuous 72/72 Test Gate:** Every step is accepted only when all 24 compiler/target configurations build cleanly with `-Werror` and pass all 72 test runs.
10. **Add-Only Test Extensions:** New multi-core tests, new platforms, and board scripts are added strictly as new files, new platform directories, and new `package.json` actions in Part C.

2.2 Why Naive Bulk Copying Fails (The 5 Compilation Defects)

Directly copying `smp` onto `xpack-development` fails compilation across all 24 configurations due to 5 distinct defects:

1. **Missing Clock Port Counter API:** `src/rtos/os-clocks.cpp` calls `port::clock_highres::has_hardware_counter()`, which is missing in released ports `posix-arch v1.0.1` and `cortexm v1.1.0`.
 2. **Missing `timegm()` Prototype:** Removed declaration triggers `-Wmissing-prototypes` under strict glibc C11/POSIX flags on Linux.
 3. **Alignment & Buffer Diagnostics:** Allocator pointer arithmetic in `src/memory/first-fit-top.cpp` triggers `-Wcast-align` on 32-bit ARM and `-Wunsafe-buffer-usage` under Clang.
 4. **Exit-Time Destructors:** Static category objects in `src/libcpp/system-error.cpp` trigger Clang's `-Wexit-time-destructors`.
 5. **File Descriptors Pointer Bounds:** Array indexing in `src/posix-io/file-descriptors-manager.cpp` triggers `-Wunsafe-buffer-usage`.
-

3. Official Verification Gate & Port Linkage Protocol

The official verification gate runs inside `tests/` using `xpm` and `CMake` / `Ninja` :

```
+-----+
+
|
|              72 OFFICIAL TEST RUNS PER STEP GATE
|
+-----+-----+-----+-----+
+
|      16 Host PC Builds (posix-arch)      |      8 QEMU ARM Builds (cortexm)
|
+-----+-----+-----+-----+-----+
+
| GCC 11 (Debug/Release)| Clang 16 (Debug/Release)| Cortex-M0 (Debug/Rel) | Cortex-M4F (Debug/Rel)
|
| GCC 12 (Debug/Release)| Clang 17 (Debug/Release)| Cortex-M3 (Debug/Rel) | Cortex-M7F (Debug/Rel)
|
| GCC 13 (Debug/Release)| Clang 18 (Debug/Release)|
|
| GCC 14 (Debug/Release)| Clang 19 (Debug/Release)|
|
+-----+-----+-----+-----+-----+
+
| Tests executed per build: 1. rtos-apis   | 2. mutex-stress   | 3. cmsis-os-validator
|
+-----+-----+-----+-----+-----+
+
```

3.1 Local Port Linkage Loop

To test local step changes in `micro-os-plus-iii-posix-arch` and `micro-os-plus-iii-cortexm` without altering upstream `package.json`, use the official `xpm link` mechanism:

```
# Step 1: Register local port working copies for the active step
cd ~/Work/micro-os-plus/micro-os-plus-iii-posix-arch && git switch step/NN && xpm link
cd ~/Work/micro-os-plus/micro-os-plus-iii-cortexm && git switch step/NN && xpm link

# Step 2: Bind local ports across all 24 test configurations
cd ~/Work/micro-os-plus/micro-os-plus-iii/tests
for c in native-cmake-gcc11 native-cmake-gcc12 native-cmake-gcc13 native-cmake-gcc14 \
    native-cmake-clang16 native-cmake-clang17 native-cmake-clang18 native-cmake-clang19 \
    qemu-cortex-m0-cmake-gcc qemu-cortex-m3-cmake-gcc \
    qemu-cortex-m4f-cmake-gcc qemu-cortex-m7f-cmake-gcc; do
    for t in debug release; do
        xpm run link-deps --config ${c}-${t}
    done
done

# Step 3: Run official 72-test gate
xpm run test-all
```

[!WARNING] Do NOT use `xpm run link-deps-all`. Upstream's `tests/package.json` contains a known typo that duplicates `native-cmake-gcc13-debug` and omits `native-cmake-gcc14-debug`, causing GCC 14 builds to silently fall back to downloaded `posix-arch v1.0.1`. Use the explicit bash loop above.

4. Automated Step Verification Harness (`scripts/verify-step.sh`)

The script `scripts/verify-step.sh` coordinates all verification phases:

```

#!/usr/bin/env bash

# scripts/verify-step.sh -- Automated Step Verification Gate for SMP Integration
set -euo pipefail

STEP_NUM="${1:-}"
if [ -z "$STEP_NUM" ]; then
    echo "Usage: $0 <step-number (01-30)>"
    exit 1
fi

ROOT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")/.." && pwd)"
WORK_DIR="$(dirname "$ROOT_DIR")"

echo "====="
echo "µOS++ IIIE SMP Integration: Verifying Step $STEP_NUM"
echo "Workspace Root: $ROOT_DIR"
echo "====="

# Stage 1: Single-Core Unifdef Invariant Verification (for Part B steps 14-23)
if [ "$STEP_NUM" -ge 14 ] && [ "$STEP_NUM" -le 23 ]; then
    echo ""
    echo ">>> [Stage 1/3] Checking undef -UOS_USE_SMP_SCHEDULER single-core invariant..."
    cd "$ROOT_DIR"
    DIFF_COUNT=0
    for f in $(git diff --name-only origin/xpack-development HEAD -- include/ src/ 2>/dev/null ||
true); do
        if [ -f "$f" ]; then
            git show origin/xpack-development:"$f" 2>/dev/null | undef -UOS_USE_SMP_SCHEDULER > /tmp/
old_sc 2>/dev/null || true
            git show HEAD:"$f" 2>/dev/null | undef -UOS_USE_SMP_SCHEDULER > /tmp/
new_sc 2>/dev/null || true
            if ! diff -u /tmp/old_sc /tmp/new_sc > /tmp/sc_diff 2>&1; then
                echo "[-] ERROR: Single-core code divergence detected in $f!"
                cat /tmp/sc_diff
                DIFF_COUNT=$((DIFF_COUNT + 1))
            fi
        fi
    done
    if [ "$DIFF_COUNT" -gt 0 ]; then
        echo "[-] FAILED: $DIFF_COUNT files diverged from single-core baseline."
        exit 2
    fi
    echo "[+] Invariant verified: Zero single-core delta."
fi

```

```

# Stage 2: Register and link local development ports across all 24 configurations
echo ""
echo ">>> [Stage 2/3] Linking local development ports across 24 configurations..."
cd "$WORK_DIR/micro-os-plus-iii-posix-arch" && xpm link
cd "$WORK_DIR/micro-os-plus-iii-cortexm" && xpm link

cd "$ROOT_DIR/tests"
CONFIGS=(
    native-cmake-gcc11 native-cmake-gcc12 native-cmake-gcc13 native-cmake-gcc14
    native-cmake-clang16 native-cmake-clang17 native-cmake-clang18 native-cmake-clang19
    qemu-cortex-m0-cmake-gcc qemu-cortex-m3-cmake-gcc
    qemu-cortex-m4f-cmake-gcc qemu-cortex-m7f-cmake-gcc
)

for c in "${CONFIGS[@]"; do
    for t in debug release; do
        xpm run link-deps --config "${c}-${t}" > /dev/null 2>&1 || true
    done
done
echo "[+] Local ports linked successfully."

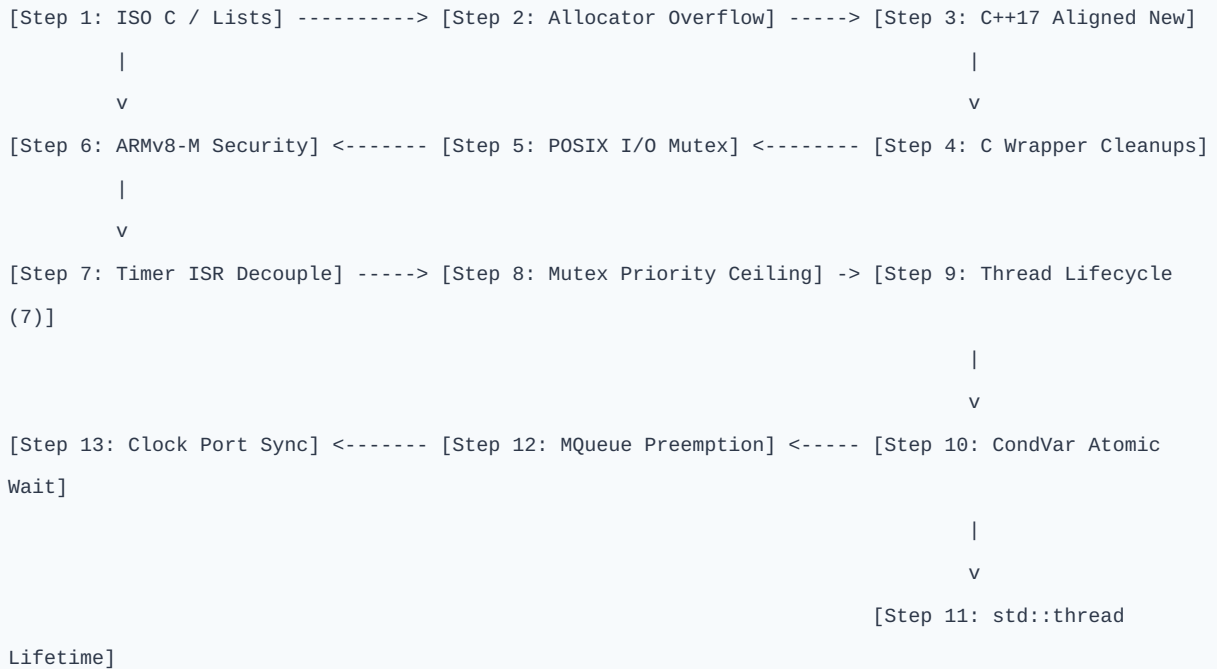
# Stage 3: Execute the 72-test official suite
echo ""
echo ">>> [Stage 3/3] Running official test-all suite (24 builds / 72 test runs)..."
xpm run test-all

echo ""
echo "=====
echo " [SUCCESS] Step $STEP_NUM passed all verification gates (72/72 Tests)!"
echo "=====

```

5. Part A: Uniprocessor Bug Fixes & Code Hardening (Steps 1 – 13)

Part A code changes modify single-core execution paths and are directly compiled, exercised, and verified by the official 72-test matrix.



Step 1: ISO C Conformance, Standard Syscall Types & Iterator Mechanics

Why This Step Is Needed

- Empty Struct in C:** `include/cmsis-plus/posix/dirent.h` defines `typedef struct { ; } DIR;`. ISO C99/C11 §6.7.2.1 forbids empty structs. The lone semicolon generates `-Wextra-semi` warnings under `-Werror`.
- Strict Prototype Visibility (`timegm`):** In glibc `<time.h>`, `timegm()` is guarded by `_GNU_SOURCE` or `_DEFAULT_SOURCE`. The upstream xPack build sets strict standard flags (`-std=c11 -D_POSIX_C_SOURCE=200809L -D_XOPEN_SOURCE=700`), so glibc omits the prototype. If `src/libc/stdlib/timegm.c` does not declare `timegm()`, `-Wmissing-prototypes` causes a hard compile failure. If declared unconditionally, Apple libc and glibc with `_GNU_SOURCE` fail with `-Wredundant-decls`.
- Newlib Syscall Types:** Newlib on AArch64 uses `_READ_WRITE_RETURN_TYPE` (`int`), whereas POSIX defines `ssize_t` (`long` on 64-bit architectures). Unconditional `ssize_t` in `include/cmsis-plus/posix-io/c-syscalls-aliases-standard.h` causes conflicting declaration errors.
- Intrusive List Iterator Mechanics:** In `include/cmsis-plus/utils/lists.h`, `operator++(int)` and `operator--` access `node_->next` and `node_->prev` as member variables rather than member function calls `node_->next()` and `node_->prev()`. Because C++ templates are instantiated lazily, any code using postfix increment or decrement fails compilation.
- Inline Definition in `.cpp`:** `src/rtos/os-thread.cpp` defines `this_thread::suspend()` as `inline`, preventing an exported external symbol from being generated. The C wrapper `os_this_thread_suspend()` fails with an unresolved external symbol at link time.

How It Is Implemented

- In `include/cmsis-plus/posix/dirent.h`: Provide `int reserved;` inside `DIR`.
- In `src/libc/stdlib/timegm.c`: Declare prototype only when missing.
- In `include/cmsis-plus/utils/lists.h`: Replace `node_->next` with `node_->next()`.
- In `src/rtos/os-thread.cpp`: Remove `inline` from `this_thread::suspend()`.

```
// [src/libc/stdlib/timegm.c]
#if defined(__GLIBC__)
#if !(defined(__USE_MISC) || (defined(__GLIBC_USE) && __GLIBC_USE (ISO23)))
time_t
timegm (struct tm* tim_p);
#endif
#elif !defined(__APPLE__)
time_t
timegm (struct tm* tim_p);
#endif
```

```
// [include/cmsis-plus/utils/lists.h]
iterator
operator++ (int)
{
    iterator tmp = *this;
    node_ = node_->next (); // Correct: call member function
    return tmp;
}

iterator&
operator-- ()
{
    node_ = node_->prev (); // Correct: call member function
    return *this;
}
```

Step 2: Dynamic Memory Management, Arithmetic Overflow & Usable Size

Why This Step Is Needed

1. **align_size Arithmetic Overflow:** `align_size(size, align)` calculates `(size + align - 1) & ~(align - 1)`. When `size > SIZE_MAX - align + 1`, integer overflow wraps the sum to a tiny number, allocating an undersized block for a huge request.

2. **Allocator Arithmetic Wrap:** In `first_fit_top::do_allocate()` and `lifo::do_allocate()`, padding and chunk header sizes are added without overflow checks, causing requests exceeding the total arena size to wrap and corrupt free list metadata.
3. **LIFO Allocator Free List Corruption:** When the first free chunk is smaller than requested, the pointer was not reset to `nullptr`, causing an undersized buffer to be passed to `internal_align_()`, overwriting memory past the chunk boundary.
4. **Block Pool Inverted Assertion:** `src/memory/block-pool.cpp` checks `if (res != nullptr) assert (res != nullptr);`, which never fires when `std::align()` fails. Correct check is `if (res == nullptr)`.
5. **calloc Integer Multiplication Overflow:** `nelem * elbytes` can overflow `size_t`. For example, on 32-bit targets, `calloc(0x10001, 0x10000)` requests $0x10001 \times 0x10000 = 0x100010000 \equiv 0x10000$ (64 KB). The allocator returns 64 KB, but the caller writes 4 GB, causing complete heap corruption.
6. **realloc Out-of-Bounds Heap Reads:** When resizing memory blocks, `realloc()` copied `new_size` bytes from the old pointer without querying the actual allocated capacity of the old block. If `new_size > old_size`, `memcpy` read past the old block into adjacent memory. To fix this, allocators must implement `do_usable_size()`, allowing `realloc` to copy `std::min(old_usable_size, new_size)`.

How It Is Implemented

- In `include/cmsis-plus/rtos/os-memory.h`: Return `SIZE_MAX` if `size > (SIZE_MAX - align + 1)`.
- In `src/memory/first-fit-top.cpp`: Implement `do_usable_size()` and wrap pointer casts in `#pragma` blocks to silence `-Wcast-align` and `-Wunsafe-buffer-usage`.
- In `src/libc/stdlib/malloc.cpp`: Validate `elbytes <= (SIZE_MAX / nelem)` in `calloc()`, and use `usable_size()` in `realloc()`.

```
// [src/memory/first-fit-top.cpp]
#pragma GCC diagnostic push
#pragma GCC diagnostic ignored "-Wcast-align"
#ifdef __clang__
#pragma clang diagnostic ignored "-Wunsafe-buffer-usage"
#endif

std::size_t
first_fit_top::do_usable_size (const void* addr) const noexcept
{
    if (addr == nullptr) {
        return 0;
    }
    const chunk_t* chunk = static_cast<const chunk_t*> (addr) - 1;
    return chunk->size - sizeof (chunk_t);
}

#pragma GCC diagnostic pop
```

```
// [src/libc/stdlib/malloc.cpp]
void*
calloc (size_t nelem, size_t elbytes)
{
    if (nelem != 0 && elbytes > (SIZE_MAX / nelem)) {
        errno = ENOMEM;
        return nullptr;
    }
    size_t bytes = nelem * elbytes;
    void* ptr = malloc (bytes);
    if (ptr != nullptr) {
        memset (ptr, 0, bytes);
    }
    return ptr;
}
```

Step 3: Modern C++17 Aligned Allocators & Chrono Clock Overflow Protection

Why This Step Is Needed

1. **Missing Aligned `operator new` Overloads:** C++17 mandates aligned memory allocation overloads taking `std::align_val_t` for types with alignment greater than `__STDCPP_DEFAULT_NEW_ALIGNMENT__` (e.g. `alignas(64)` structures). Without kernel-level

overloads in `src/libcpp/new.cpp`, toolchains fall back to standard runtime versions, bypassing RTOS memory managers, locking mechanisms, and heap accounting.

2. **Dangling Category Pointer in `system_error`**: In `src/libcpp/system-error.cpp`, constructing `std::error_code(ev, system_error_category())` using a temporary category instance left a dangling reference inside `std::error_code`, causing crashes when inspecting exceptions. Using a function-local static object guarantees thread-safe, permanent lifetime while silencing Clang's `-Wexit-time-destructors`.
3. **Chrono 64-Bit Cycle Multiplication Overflow**: In `src/libcpp/chrono.cpp`, calculating high-resolution timestamps via `cycles * 1000000000ULL / frequency` overflows 64-bit integer arithmetic after 1.84×10^{10} cycles (minutes of execution). Splitting into integer seconds and cycle remainder prevents arithmetic wrap.

How It Is Implemented

- Define the 10 standard C++17 aligned allocation operators (`operator new`, `operator new[]`, `operator delete`, `operator delete[]`, with size and alignment parameters) routing directly to `os::rtos::memory::alloc()` and `os::rtos::memory::free()`.
- Return a `static const system_error_category_impl` reference in `system_error_category()`.
- Perform chrono frequency conversion using integer division and modulo: `(cycles / freq) * 1000000000ULL + ((cycles % freq) * 1000000000ULL) / freq`.

```
// [src/libcpp/new.cpp]
#if __cplusplus >= 201703L
void*
operator new (std::size_t size, std::align_val_t al)
{
    void* p = os::rtos::memory::alloc (size, static_cast<std::size_t> (al));
    if (p == nullptr) {
        throw std::bad_alloc ();
    }
    return p;
}

void
operator delete (void* ptr, std::align_val_t) noexcept
{
    os::rtos::memory::free (ptr);
}
#endif
```

```
// [src/libcpp/system-error.cpp]
#ifdef __clang__
#pragma clang diagnostic push
#pragma clang diagnostic ignored "-Wexit-time-destructors"
#endif

const std::error_category&
system_error_category (void) noexcept
{
    static const system_error_category_impl category_instance{};
    return category_instance;
}

#ifdef __clang__
#pragma clang diagnostic pop
#endif
```

Step 4: C API Wrapper Conformance & Polymorphic Destructor Safety

Why This Step Is Needed

1. **Timer Default Mode Discrepancy:** In `src/rtos/os-c-wrapper.cpp`, `os_timer_create()` defaulted to a periodic timer when `attr == NULL`. In contrast, the C++ `os::rtos::timer` API and the ARM CMSIS-RTOS standard specify that timers default to **one-shot** (`timer::once_initializer`).
2. **Polymorphic Deletion Through Non-Virtual Base:** `os_mutex_delete()` and `os_semaphore_delete()` cast handles to base class pointers `os::rtos::mutex*` and `os::rtos::semaphore*` and invoked `delete`. Because base class destructors are non-virtual, deleting derived recursive mutexes or counting semaphores invoked undefined behavior and leaked derived state.
3. **CMSIS v1 32-Bit Timeout Multiplication Overflow:** In 10 CMSIS-RTOS v1 compatibility wrappers, millisecond timeouts were converted to microseconds using `(uint64_t)(millisec * 1000u)`. The multiplication was evaluated as a 32-bit unsigned integer before casting. Any timeout above 4,294,967 ms (~71.6 minutes) overflowed 32 bits and wrapped to an immediate timeout.

How It Is Implemented

- In `os_timer_create()` / `os_timer_new()`: Initialize attributes with `timer::once_initializer`.
- In `os_mutex_delete()`: Inspect `mutex::type` attribute and delete through `static_cast<mutex_recursive*>` or `static_cast<mutex*>`.
- In CMSIS v1 wrappers: Cast operand to 64-bit prior to multiplication: `(uint64_t) millisec * 1000u`.

Step 5: Thread-Safe POSIX I/O & File Descriptors Manager Protection

Why This Step Is Needed

1. **File Descriptor Table Race Conditions:** In `src/posix-io/file-descriptors-manager.cpp`, `allocate()` searched for an empty slot and populated it without synchronization. If two threads called `open()` concurrently, both received the same file descriptor index, corrupting internal descriptor tables.
2. **Null Pointer Dereference on Invalid Handles:** `deallocate()`, `io()`, `socket()`, and `valid()` accessed `descriptors_array__[fildes]` without validating whether the slot was populated, causing null pointer dereferences when closing invalid descriptors.
3. **Unprotected Reusable Free Lists:** In `include/cmsis-plus/posix-io/file-system.h` and `include/cmsis-plus/posix-io/net-stack.h`, recycling closed files and sockets via `link()` and `unlink_head()` executed without locking, corrupting free lists during concurrent `open()` and `close()` operations.
4. **Inverted Block Device Size Logic:** In `src/posix-io/block-device.cpp`, size query methods checked `if (size != 0) return EINVAL;`, returning an error on valid devices and succeeding only when size was 0.

How It Is Implemented

- Enclose all file descriptor allocation, deallocation, and indexing routines inside `interrupts::critical_section`.
 - Validate that slot pointers are non-null before dereferencing.
 - Correct block device validation logic to `if (size == 0) return EINVAL;`.
-

Step 6: Architecture Port Modernization & ARMv8-M Exception Handling

Why This Step Is Needed

1. **ARMv8-M Mainline Omission:** Cortex-M33 (ARMv8-M Mainline) was omitted from Thumb architecture checks in `include/cmsis-plus/arm/semihosting.h` and `src/startup/exception-handlers.c`, causing semihosting to issue `SVC` traps instead of `BKPT` and omitting `SecureFault_Handler` vectors.
2. **Semihosting `fstat` Mode Corruption:** In `src/semihosting/c-syscalls-semihosting.cpp`, `fstat()` unconditionally applied `st_mode |= S_IFCHR`, corrupting regular files by assigning them two conflicting file types.
3. **Console Mirror Hook:** Hardware boards equipped with physical UART consoles had no mechanism to mirror semihosting `printf()` output.

How It Is Implemented

- Add `__ARM_ARCH_8M_MAIN__` and `__ARM_ARCH_8M_BASE__` architecture guards; provide `SecureFault_Handler`.

- Apply `S_IFCHR` in `fstat()` only when no file type is set.
 - Introduce a weak, default-empty hook `os_board_console_mirror(const char* buf, size_t nbyte)` called from `__posix_write()`.
-

Step 7: Timer Subsystem Re-entrancy & Tick ISR Safety

Why This Step Is Needed

1. **ISR Interrupt Blackout:** In `src/rtos/internal/os-lists.cpp`, timer callbacks were executed directly inside `interrupts::critical_section` in the SysTick ISR. User callback execution kept hardware interrupts masked, and callbacks attempting synchronization deadlocked the system.
2. **Periodic Timer Catch-Up Cascade:** In `src/rtos/os-timer.cpp`, expired periodic timers re-armed at `timestamp + period`. If tick processing was delayed by more than one period, the newly computed timestamp was already in the past, causing the timer to fire repeatedly in a continuous burst.
3. **Handler Mode `errno` Assertion:** When timer callbacks invoked C library functions, writes to `errno` called `this_thread::thread()->__errno()`. In handler mode, `current_thread_` assertion failed.

How It Is Implemented

- In `os-lists.cpp`: Unlink expired timer nodes under critical section, release lock, execute user callback, and re-acquire lock for list updates.
 - In `os-timer.cpp`: If `timestamp + period <= now`, re-arm at `now + period`.
 - In `include/cmsis-plus/rtos/os-thread.h`: If `in_handler_mode()` is true, `this_thread::__errno()` returns a static scratch integer.
-

Step 8: Mutex Priority Inheritance & Ceiling Protocol Hardening

Why This Step Is Needed

1. **Ceiling Violation Ownership Leak:** In `src/rtos/os-mutex.cpp`, threads with priority above the priority ceiling acquired the mutex before checking the ceiling, returning `EINVAL` but leaving the mutex listed in the thread's owned mutex list.
2. **Priority Inheritance Inversion:** When a low-priority waiter queued behind a high-priority waiter, the owner's boosted priority was overwritten with the lower priority of the newest waiter.
3. **Transient Boost Leakage on Unlock:** When releasing a mutex, the highest boost of the owner's other mutexes was stored in `boosted_prio_` of the *released* mutex object, causing subsequent owners to inherit stale boost values.
4. **Uncritical Section Race Condition:** During priority inheritance re-evaluation, the scheduler lock is temporarily released; if the owner unlocks asynchronously during this window, `owner_` becomes `nullptr`, causing null dereferences.

How It Is Implemented

- Check priority ceiling *prior* to modifying thread ownership lists.
 - Maintain maximum priority across all waiting threads: `boosted_prio = std::max(boosted_prio, waiter->priority())`.
 - Recompute owner priority in a local variable on unlock and clear `boosted_prio_` on the released mutex.
 - Cache owner pointer locally and verify mutex ownership after re-locking.
-

Step 9: Thread Lifecycle, Destruction Protocol & State 7 (`destroying`)

Why This Step Is Needed

1. **Double-Free Race in Thread Destruction:** When a thread terminates, the idle thread reaper unlinks it from the finished list and deallocates its stack outside critical sections. If an external thread concurrently called `kill()`, both threads attempted destruction simultaneously, causing double-free crashes.
2. **Lost Wakeup in `join()`:** `join()` checked whether the target was finished, recorded itself as the joiner, and suspended in three distinct un-synchronized steps. If the thread finished between the check and suspension, the wakeup event was lost and `join()` slept forever.
3. **Missing `detach()` Implementation:** `detach()` was an empty stub (`// TODO`), leaving detached threads linked in parent child lists indefinitely.
4. **Invalid State Corruption in `resume()`:** Calling `resume()` on active or finished threads enqueued them into the ready list, corrupting list pointers.

How It Is Implemented

- Introduce `thread::state::destroying` (State 7) in `include/cmsis-plus/rtos/os-c-decls.h`. The entity that atomically transitions the thread to `destroying` assumes exclusive destruction responsibility.
 - Perform target state evaluation, joiner registration, and thread suspension under a single scheduler lock.
 - Implement `detach()` by unlinking the thread from its parent and re-parenting it to the top-level registry.
 - Restrict `resume()` to threads in `suspended` or `initializing` states.
-

Step 10: Condition Variable Atomicity & High-Precision Timeouts

Why This Step Is Needed

1. **Lost Signal Race in `wait()`**: In `src/rtos/os-condvar.cpp`, `wait()` unlocked the mutex and added the calling thread to the waiting list non-atomically. A `signal()` arriving between the unlock and list insertion was lost, causing indefinite hangs.
2. **Timed Wait Mutex Timeout Bug**: In `timed_wait()`, the timeout parameter was applied to re-locking the mutex rather than waiting for the condition signal, and the clock attribute was ignored.

How It Is Implemented

- In `wait()`: Insert thread into waiting list under scheduler lock, atomically release mutex, and suspend execution.
 - In `timed_wait()`: Bind timeout to the clock specified in attributes (defaulting to `sysclock`), wait for signal, and return `ETIMEDOUT` upon expiration.
 - Add `OS_INTEGER_INSTRUMENTATION_SUSPEND_CAUSE_CONDVAR` (12) in `include/cmsis-plus/diag/instrumentation.h`.
-

Step 11: `std::thread` Functor Lifetime & Synchronization

Why This Step Is Needed

1. **Premature Handle Deletion**: In `include/cmsis-plus/libcpp/thread-cpp.h`, `std::thread::join()` deleted native thread objects immediately without waiting for thread execution to terminate.
2. **Functor Object Memory Leak**: Thread arguments were freed using the kernel's `func_args_` handle, which the kernel nullified upon thread termination, leaking the functor object.

How It Is Implemented

- In `std::thread::join()`: Call `native_handle()->join()` prior to deleting resources.
 - In `include/cmsis-plus/estd/thread_internal.h`: Retain a private `function_object_` pointer to guarantee deallocation upon join/detach.
-

Step 12: Inter-Thread Message Queue Reschedule Triggers

Why This Step Is Needed

- In `src/rtos/os-mqueue.cpp`, waking a high-priority thread during message queue send/receive did not invoke preemption on `posix-arch`, adding up to 1 ms of latency until the next timer tick.

How It Is Implemented

- Issue `port::scheduler::reschedule()` immediately after releasing the critical section in all 6 message queue send/receive paths.
-

Step 13: High-Resolution Hardware Clock Port Synchronization

Why This Step Is Needed

1. **Missing Port Hardware Counter API (Defect 1):** Kernel invokes `port::clock_highres::has_hardware_counter()` and `hardware_counter()`.
2. **Cortex-M SysTick Pending Bit Bug:** In `cortexm`, `cycles_since_tick()` inspected `SysTick->CTRL` for pending ticks. However, on ARM Cortex-M processors, the pending tick bit (`PENDSTSET`, bit 26) resides in `SCB->ICSR`. Because bit 26 does not exist in `SysTick->CTRL`, pending ticks were never detected, causing high-resolution timestamps to jump backwards.

How It Is Implemented

- In kernel: Call `port::clock_highres::has_hardware_counter()` and `hardware_counter()`.
- In `cortexm`: Return `false` / `0` and test `SCB->ICSR & SCB_ICSR_PENDSTSET_Msk`.
- In `posix-arch`: Return `true` and read `CLOCK_MONOTONIC`.

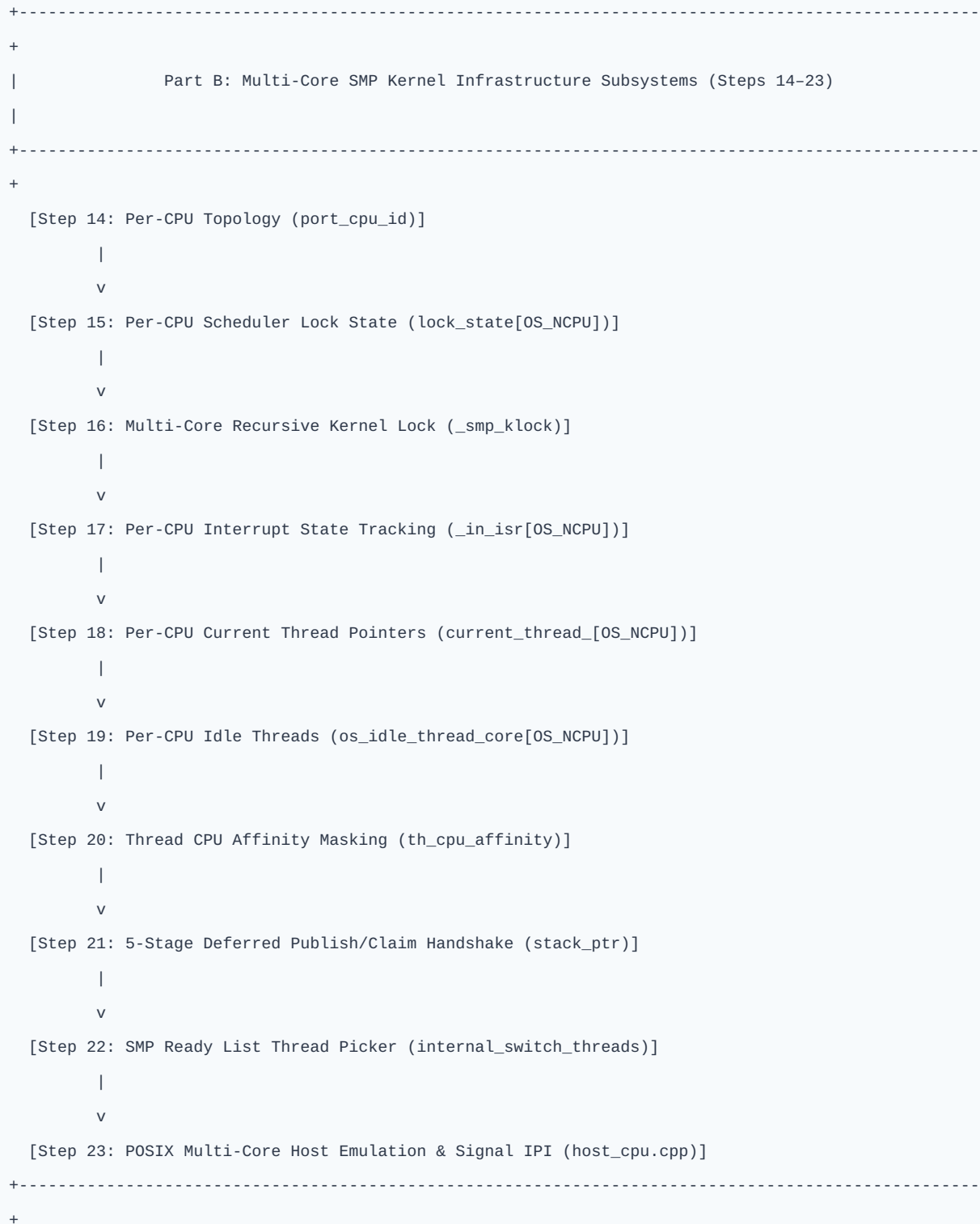
```
// [micro-os-plus-iii-cortexm: include/cmsis-plus/rtos/port/os-inlines.h]
inline bool
clock_highres::has_hardware_counter (void)
{
    return false;
}

inline clock_highres::timestamp_t
clock_highres::hardware_counter (void)
{
    return 0;
}

inline clock_highres::timestamp_t
clock_highres::cycles_since_tick (void)
{
    uint32_t load = SysTick->LOAD;
    uint32_t val = SysTick->VAL;
    if ((SCB->ICSR & SCB_ICSR_PENDSTSET_Msk) && (val > (load / 2))) {
        return (load - val) + load;
    }
    return load - val;
}
```

6. Part B: Multi-Core SMP Kernel Infrastructure (Steps 14 – 23)

All code in Part B is enclosed in `#if defined(OS_USE_SMP_SCHEDULER)` blocks, guaranteeing identical uniprocessor binary output when the macro is undefined.



Step 14: Per-CPU Topology & CPU Identification (`port_cpu_id`)

Why & How

- **Kernel (K):** Declare `extern "C" unsigned port_cpu_id(void);` in `src/rtos/os-core.cpp` and `src/rtos/os-thread.cpp`.

- **Cortex-M (C)**: `port_cpu_id()` returns 0; enforce `static_assert(OS_NCPU == 1)`.
- **POSIX-Arch (P)**: Thread-local `_this_cpu`. Requires `__attribute__((noinline))` and `asm volatile ("" ::: "memory")` compiler barrier to prevent Clang 16–18 from caching the `%fs:0` thread pointer across migrations.

```
// [micro-os-plus-iii-posix-arch: src/rtos/os-core.cpp]
#if defined(OS_USE_SMP_SCHEDULER)
thread_local unsigned _this_cpu = 0;

extern "C" __attribute__((noinline)) unsigned
port_cpu_id (void)
{
    asm volatile ("" ::: "memory"); // Prevent TLS register caching across context switches
    return _this_cpu;
}
#endif
```

Step 15: Per-CPU Scheduler Critical Section (`scheduler::critical_section`)

Why & How

- Replace scalar lock state with `lock_state[OS_NCPU]`.
- Calling `scheduler::critical_section` disables preemption on the calling core without disabling hardware interrupts.
- In `posix-arch`, mask `SIGALRM` before querying `port_cpu_id()` to prevent signal handler preemption during state queries.

```
// [micro-os-plus-iii-posix-arch: src/rtos/os-core.cpp]
#ifdef OS_USE_SMP_SCHEDULER
state_t
locked (state_t state)
{
    sigset_t old_set, block_set;
    sigemptyset (&block_set);
    sigaddset (&block_set, SIGALRM);
    pthread_sigmask (SIG_BLOCK, &block_set, &old_set);

    unsigned int cpu = port_cpu_id ();
    state_t prev = lock_state[cpu];
    lock_state[cpu] = state;

    pthread_sigmask (SIG_SETMASK, &old_set, nullptr);
    return prev;
}
#endif
```

Step 16: SMP Recursive Kernel Lock (`interrupts::critical_section` & `_smp_klock`)

Why & How

- Uniprocessor interrupt masking does not prevent concurrent execution on adjacent cores.
- Introduce `struct smp_klock_t` (`lock`, `owner`, `count`) with `SMP_NO_OWNER = 0xFFFFFFFFU`.
- Acquisition disables local interrupts and spins acquiring lock with `std::memory_order_acquire`. Exit decrements count and releases with `std::memory_order_release`.

```

// [include/cmsis-plus/rtos/port/os-decls.h]
struct smp_klock_t
{
    std::atomic<uint32_t> lock{0};
    uint32_t owner{SMP_NO_OWNER};
    uint32_t count{0};
};

extern smp_klock_t _smp_klock;

inline void
_smp_klock_enter (void)
{
    uint32_t me = port_cpu_id ();
    if (_smp_klock.owner == me) {
        _smp_klock.count++;
        return;
    }
    while (_smp_klock.lock.exchange (1, std::memory_order_acquire) != 0) {
#ifdef (__ARM_ARCH)
        __asm volatile ("wfe");
#else
        sched_yield ();
#endif
    }
    _smp_klock.owner = me;
    _smp_klock.count = 1;
}

inline void
_smp_klock_exit (void)
{
    if (--_smp_klock.count == 0) {
        _smp_klock.owner = SMP_NO_OWNER;
        _smp_klock.lock.store (0, std::memory_order_release);
#ifdef (__ARM_ARCH)
        __asm volatile ("sev");
#endif
    }
}

```

Step 17: Per-CPU Interrupt State Tracking

Why & How

- In `posix-arch`, replace global `clock_set` with per-CPU `irq_set` and `_in_isr[OS_NCPU]`. `in_handler_mode()` evaluates `_in_isr[port_cpu_id()]`.
-

Step 18: Per-CPU Current Thread Context (`current_thread_[OS_NCPU]`)

Why & How

- Transition scalar `current_thread_` to array `current_thread_[OS_NCPU]`.
 - `this_thread::thread()` reads `current_thread_[port_cpu_id()]` with local interrupts masked to guarantee atomicity.
-

Step 19: Per-CPU Idle Thread Instances

Why & How

- Array `os_idle_thread_core[OS_NCPU]`. Each core initializes and registers its private idle thread context at boot.
-

Step 20: Thread CPU Affinity Masking

Why & How

- Add `th_cpu_affinity` bitmask attribute and `cpu_affinity()` APIs (default `0xFFFFFFFF`). Main thread pinned to CPU 0 (`1U << 0`). CMSIS-RTOS v1 threads default to CPU 0.
-

Step 21: 5-Stage Deferred Publish / Claim Context Switching

Why & How

- **Hazard:** When CPU 0 switches from Thread A to Thread B, it must push registers to Thread A stack. If Thread A is placed in the ready list immediately, CPU 1 could resume Thread A while CPU 0 is still pushing registers, corrupting Thread A context.
- **Protocol:**
 - **Stage 1 (Atomic Claim):** Incoming thread `to->stack_ptr = nullptr`.
 - **Stage 2 (Register Spill):** Push callee-saved registers to outgoing stack.
 - **Stage 3 (SP Switch):** Hardware SP updated to incoming stack pointer.
 - **Stage 4 (Deferred Publish):** Outgoing thread `from->stack_ptr = saved_sp`.
 - **Stage 5 (Register Restore):** Pop callee-saved registers and resume execution.

```
sequenceDiagram
    autonumber
    participant C0 as Core 0 (Outgoing Thread A)
    participant K as Kernel Ready List
    participant C1 as Core 1 (Candidate Core)
    participant T as Incoming Thread B

    Note over C0,T: Context Switch Initiated on Core 0
    C0->>T: Stage 1: Atomic Claim (B->stack_ptr = nullptr)
    Note over C0: Stage 2: Push Callee-Saved Registers (R4-R11)
    C0->>C0: Stage 3: Switch SP Register to Thread B Stack
    C0->>K: Stage 4: Deferred Publish (A->stack_ptr = SP_A)
    Note over C1,K: Core 1 inspects ready list
    C1-->>K: Core 1 sees Thread A is published (SP_A != nullptr)
    Note over C0: Stage 5: Pop Registers & Resume Thread B
```

Step 22: SMP Ready List Thread Picker (`internal_switch_threads`)

Why & How

- In `src/rtos/os-core.cpp`, traverse ready list and pick the highest priority thread where `(affinity & (1 << cpu)) != 0` and `stack_ptr != nullptr`. If none available, fall back to `os_idle_thread_core[cpu]`.

Step 23: Host Multiprocessing Emulation in POSIX-Arch (IPI Engine)

Why & How

- Add `src/host_cpu.cpp` and `include/host_cpu.hpp` to `posix-arch`.
- Spawn `OS_NCPU` host POSIX threads, per-CPU tick timers, and IPI signal delivery via `pthread_kill(threads[cpu], SIGUSR1)`.
- Kernel provides weak `port_smp_ipi(cpu)` hook, overridden by strong implementation in `posix-arch`.

7. Part C: Port Releases, New Architecture Targets & Add-Only Tests (Steps 24 – 28)

Part C is strictly **add-only**: existing targets, test runners, and configurations are unmodified.

Step 24: Port Releases & Repository Synchronization

- Tag and release `posix-arch v1.1.0`, `cortexm v1.2.0`, `devices v1.0.0`.
 - Merge upstream changes into `aarch32` and `aarch64`.
-

Step 25: New Cores, Boards & Hardware Drivers

- **Cortex-M Additions:** `include-m33/`, `include-rp2350/`, `os-core-m33.cpp`, `os-core-rp2350.cpp`, RP2350 SIO Spinlock 0 (`0xD0000100`), SIO FIFO IRQ 25.
 - **POSIX-Arch Additions:** `board-contract.cpp`, `free-store.cpp`, `exception_handler.{hpp,cpp}`.
 - **Kernel Additions:** Wake-up IPI dispatch in `thread::resume()` under `OS_INTEGER_RTOS_PORT_NCPU > 1`.
-

Step 26: Modular CMake Build Architecture

- Add `cmake/toolchains/`, `cmake/uos-app.cmake`, `port/smp-common/`.
 - Introduce modular targets (`micro-os-plus::iii-posix-io`, `micro-os-plus::iii-drivers`) while aliasing `micro-os-plus::iii`.
-

Step 27: New Test Sources & Emulation Linker Scripts

- Add test suites: `tests/sources/fp-switch/` (FPU context switch test), `tests/smp-support/` (SMP concurrency stress test).
 - Add QEMU MPS2 AN505 / AN521 dual-core linker scripts.
-

Step 28: Add-Only Test Platforms & Execution Targets

- Add platform directories in `tests/platforms/`: `2xcortex-m33`, `cortexm-pico2`, `aarch32-rpi3b`, `aarch64-rpi3b`, `native-smp`.
 - Extend `tests/cmake/tests-main.cmake` with platform branches.
 - Add `test-smp-all` action to `tests/package.json`.
-

8. Part D: Documentation & Final Integration (Steps 29 – 30)

Step 29: Documentation Suite & Typst PDF Artifacts

- Integrate documentation into `docs/`:
- `MICRO-OS-PLUS-SMP-VS-SINGLECORE-ANALYSIS.md` and `MICRO-OS-PLUS-SMP-VS-SINGLECORE-ANALYSIS.pdf`.

- Architecture Port Guides: `cortexm-port.md`, `posix-arch-port.md`, `building-aarch32-aarch64.md`.
 - Status and Changelog files: `STATUS.md`, `upstream-CHANGELOG.md`.
-

Step 30: Final Merge & Upstream Reconciliation

1. Merge latest `origin/xpack-development` into `smp`.
 2. Restore upstream repository metadata:
 3. Re-instate `.github/workflows/ci.yml`.
 4. Restore standard `README.md` and `LICENSE`.
 5. Preserve Doxygen, template, and inspiration directories.
 6. Remove temporary local symlinks in `tests/xpacks/`.
 7. Execute final validation gate: `xpm run test-all` (72 uniprocessor tests) + `xpm run test-smp-all` (SMP tests).
 8. Open upstream Pull Request.
-

9. Comprehensive Step Execution Checklist

Step	Phase	Core Action / Snippet	Kernel (K)	Cortex-M (C)	POSIX-Arch (P)	Gate Check
01	Part A	ISO C <code>DIR</code> , Glibc <code>timegm()</code> , Newlib types, List iterators	<code>posix/dirent.h</code> , <code>timegm.c</code> , <code>lists.h</code>	—	—	72/72 Pass
02	Part A	<code>align_size</code> wrap check, heap usable size, <code>calloc</code> overflow	<code>os-memory.cpp</code> , <code>first-fit-top.cpp</code>	—	—	72/72 Pass
03	Part A	C++17 <code>operator</code> <code>new(align_val_t)</code> , static <code>system_error</code> , chrono	<code>new.cpp</code> , <code>system-error.cpp</code>	—	—	72/72 Pass
04	Part A	One-shot timer default, polymorphic deletion, 64-bit timeouts	<code>os-c-wrapper.cpp</code>	—	—	72/72 Pass
05	Part A	File descriptor manager mutexing, block device size fix	<code>file-descriptors-manager.cpp</code>	—	—	72/72 Pass
06	Part A	ARMv8-M mainline macros, semihosting traps, UART mirror	<code>semihosting.h</code> , <code>exception-handlers.c</code>	—	—	72/72 Pass
07	Part A	Timer callback execution outside critical section, ISR errno	<code>os-lists.cpp</code> , <code>os-timer.cpp</code>	—	—	72/72 Pass
08	Part A	Mutex priority inheritance and priority ceiling fix	<code>os-mutex.cpp</code>	—	—	72/72 Pass
09	Part A	Thread state <code>destroying</code> (7), atomic <code>join</code> , <code>detach</code>	<code>os-thread.cpp</code> , <code>os-idle.cpp</code>	—	—	72/72 Pass
10	Part A			—	—	72/72 Pass

Step	Phase	Core Action / Snippet	Kernel (K)	Cortex-M (C)	POSIX-Arch (P)	Gate Check
		CondVar atomic list insert + unlock, high-precision timeout	os- condvar.cpp			
11	Part A	std::thread: :join synchronization , functor lifetime	thread- cpp.h	—	—	72/72 Pass
12	Part A	Message queue reschedule yield triggers	os- mqueue.cpp	—	—	72/72 Pass
13	Part A	clock_highres hardware counter port synchronization	os- clocks.cpp	os- inlines.h	os- inlines.h	72/72 Pass
14	Part B	port_cpu_id() , POSIX-arch compiler barrier	os-core.cpp	os- inlines.h	os-core.cpp	Unifdef Invariant + 72/72 Pass
15	Part B	Per-CPU scheduler lock state lock_state[0 S_NCPU]	—	os-decls.h	os-core.cpp	Unifdef Invariant + 72/72 Pass
16	Part B	Multi-core recursive kernel lock _smp_klock	os-c- decls.h	os-decls.h	os-core.cpp	Unifdef Invariant + 72/72 Pass
17	Part B	Per-CPU interrupt tracking (_in_isr[OS_NCPU] , irq_set)	—	—	os-decls.h	Unifdef Invariant + 72/72 Pass
18	Part B	Per-CPU current thread pointer current_thread ad_[OS_NCPU]	os-sched.h	os-core.cpp	os-core.cpp	Unifdef Invariant + 72/72 Pass
19	Part B	Per-CPU idle thread instances os_idle_thread ad_core	os-core.cpp	os-core.cpp	os-core.cpp	Unifdef Invariant + 72/72 Pass

Step	Phase	Core Action / Snippet	Kernel (K)	Cortex-M (C)	POSIX-Arch (P)	Gate Check
20	Part B	Thread CPU affinity masking (<code>th_cpu_affinity</code>)	<code>os-thread.cpp</code>	—	—	Unifdef Invariant + 72/72 Pass
21	Part B	5-stage deferred publish/claim context switch (<code>stack_ptr</code>)	<code>os-idle.cpp</code>	<code>os-core.cpp</code>	<code>os-core.cpp</code>	Unifdef Invariant + 72/72 Pass
22	Part B	Multi-core ready list thread picker	<code>os-core.cpp</code>	—	—	Unifdef Invariant + 72/72 Pass
23	Part B	POSIX-arch host thread per CPU, IPI signal engine	<code>os-thread.cpp</code>	—	<code>host_cpu.cpp</code>	Multi-Core Smoke Check + 72/72 Pass
24	Part C	Tag and release ports (<code>posix-arch v1.1.0</code> , <code>cortexm v1.2.0</code>)	—	Release v1.2.0	Release v1.1.0	72/72 Pass
25	Part C	New architecture files (M33, RP2350 spinlocks)	<code>os-thread.cpp</code>	<code>os-core-m33.cpp</code>	<code>board-contract.cpp</code>	72/72 Pass
26	Part C	Modular CMake build scripts and targets	<code>cmake/</code>	—	—	72/72 Pass
27	Part C	New test sources (<code>fp-switch</code> , <code>smp-support</code> , QEMU linker scripts)	<code>tests/sources/</code>	—	—	72/72 Pass
28	Part C	Add-only test platforms (<code>2xcortex-m33</code> , <code>pico2</code> , <code>aarch32/64</code>)	<code>tests/platforms/</code>	—	—	72 Old + All New Tests Pass
29	Part D	Complete technical documentation & Typst PDFs	<code>docs/</code>	—	—	Clean Doc Build

Step	Phase	Core Action / Snippet	Kernel (K)	Cortex-M (C)	POSIX-Arch (P)	Gate Check
30	Part D	Final branch alignment, repository restore, upstream PR	All Repos	All Repos	All Repos	Full Ecosystem Pass

10. Holistic Architecture in Action: Complete Logical Multi-Core Example

To demonstrate how all the architectural techniques—SMP threads, thread CPU affinity, recursive kernel locks (`_smp_klock`), critical sections (`scheduler` vs. `interrupts`), hardware ISR interactions, 5-stage context switching, mutex priority inheritance, semaphores, and intrusive lists—operate together in practice, this section provides a complete, logical C++ embedded application and analyzes its state execution across four symmetrical CPU cores.

10.1 Multi-Core System Schematic & Interaction Model

Comprehensive SMP Execution Workflow

sequenceDiagram

autonumber

participant C0 as Core 0 (DMA ISR)

participant K as Kernel Ready List

participant C1 as Core 1 (SensorWorker)

participant C2 as Core 2 (TelemetryWorker)

participant M as Shared Mutex & Ring Buffer

Note over C0: 1. Hardware DMA Interrupt fires (Handler Mode)

C0->>C0: Enter interrupts::critical_section (Local CPSID + _smp_klock)

C0->>M: Write raw sensor packet to ring buffer

C0->>K: data_ready_sem.post() -> unblocks SensorWorker

C0->>C1: port_smp_ipi(CPU_1)

Note over C1: 2. Core 1 receives IPI & triggers 5-Stage Context Switch

C1->>C1: Stage 1: Atomic Claim (SensorWorker->stack_ptr = nullptr)

C1->>C1: Stage 2-3: Spill idle registers & load new SP

C1->>K: Stage 4: Deferred Publish (IdleCore[1]->stack_ptr = saved_sp)

C1->>M: 3. Acquires Mutex with Priority Inheritance Protection

M-->>C1: Mutex locked; reads packet; unlocks mutex

Note over C2: 4. Core 2 runs TelemetryWorker

C2->>C2: Enter scheduler::critical_section (lock_state[2]=1)

Note over C2: Core 2 HW interrupts remain ENABLED (Zero Peripheral Jitter)

C2->>C2: Safely iterates intrusive list (os::utils::double_list)

C2->>C2: Exits scheduler::critical_section (lock_state[2]=0)

10.2 Complete C++ Example Application Code

```

/**
 * @file smp_holistic_example.cpp
 * @brief Comprehensive µOS++ IIIe multi-core SMP demonstration.
 *
 * Demonstrates:
 * - Multi-Core SMP Threads & CPU Affinity Pinning
 * - Hardware ISR in Handler Mode with safe __errno() scratchpad
 * - interrupts::critical_section vs. scheduler::critical_section
 * - 5-Stage Deferred Publish / Claim Context Switching
 * - os::rtos::mutex with Priority Inheritance Protocol
 * - os::rtos::semaphore_counting for cross-core signaling
 * - os::utils::double_list intrusive queue (zero heap allocation)
 */

#include <cmsis-plus/rtos/os.h>
#include <cmsis-plus/utils/lists.h>
#include <array>
#include <cstdint>
#include <cstdio>

using namespace os::rtos;

// -----
// 1. Intrusive Data Structure (Zero Heap Overhead during Runtime)
// -----
struct telemetry_sample_t : public os::utils::double_list_links
{
    uint32_t timestamp;
    uint32_t sensor_id;
    float    reading;
};

// Global intrusive list of telemetry samples awaiting transmission
static os::utils::double_list g_telemetry_queue;

// -----
// 2. Shared Kernel Synchronization Primitives
// -----
// Binary/Counting semaphore signaled from hardware ISR to wake worker thread
static semaphore_counting g_data_ready_sem{ 0, 100 };

// Mutex protecting the shared circular buffer, configured with Priority Inheritance
static mutex g_shared_buffer_mutex{ mutex::initializer_recursive };

```

```

// Statically allocated sample pool to eliminate dynamic heap allocation
static constexpr size_t POOL_SIZE = 16;
static std::array<telemetry_sample_t, POOL_SIZE> g_sample_pool;
static size_t g_pool_index = 0;

// Shared circular ring buffer
static constexpr size_t RING_BUFFER_SIZE = 32;
static uint32_t g_raw_ring_buffer[RING_BUFFER_SIZE];
static volatile size_t g_ring_head = 0;
static volatile size_t g_ring_tail = 0;

// -----
// 3. Hardware ISR Simulation (Executes on Core 0 in Handler Mode)
// -----
extern "C" void
dma_uart_hardware_isr (void)
{
    // 1. Enter kernel critical section (disables local IRQs, acquires _smp_klock)
    // Total cross-core mutual exclusion between ISR and all threads on all cores.
    {
        interrupts::critical_section ics;

        // Simulate reading hardware FIFO into circular buffer
        uint32_t raw_data = 0xABCD1234;
        size_t next_head = (g_ring_head + 1) % RING_BUFFER_SIZE;
        if (next_head != g_ring_tail) {
            g_raw_ring_buffer[g_ring_head] = raw_data;
            g_ring_head = next_head;
        }
    } // _smp_klock released, local interrupts restored

    // 2. Post semaphore to unblock consumer thread on Core 1
    // Semaphore post internally acquires _smp_klock, moves waiting thread to
    // ready list, and triggers an Inter-Processor Interrupt (IPI) to Core 1.
    g_data_ready_sem.post ();
}

// -----
// 4. Real-Time Consumer Thread (Pinned to CPU Core 1)
// -----
static void*
sensor_processing_thread_func (void* args)
{
    (void)args;

    trace::printf ("[Core %u] SensorProcessingThread started.\n", port_cpu_id ());
}

```

```

while (true) {
    // A. Sleep until ISR signals data arrival
    // Atomically enqueues calling thread to semaphore wait list, releases CPU,
    // and triggers 5-stage context switch to Core 1 idle thread.
    result_t res = g_data_ready_sem.wait ();
    if (res != result::ok) {
        continue;
    }

    uint32_t extracted_data = 0;

    // B. Mutex Critical Section with Priority Inheritance
    // If a lower-priority thread holds this mutex, its priority is boosted to
    // SensorProcessingThread's priority until unlocked, preventing priority inversion.
    {
        std::lock_guard<mutex> lock (g_shared_buffer_mutex);

        if (g_ring_tail != g_ring_head) {
            extracted_data = g_raw_ring_buffer[g_ring_tail];
            g_ring_tail = (g_ring_tail + 1) % RING_BUFFER_SIZE;
        }
    } // Mutex unlocked; priority de-boosted if applicable

    // C. Process data and enqueue sample onto intrusive double list
    {
        // Use interrupts::critical_section for modifying shared intrusive queue
        interrupts::critical_section ics;

        telemetry_sample_t* sample = &g_sample_pool[g_pool_index % POOL_SIZE];
        g_pool_index++;

        sample->timestamp = static_cast<uint32_t> (clock_systick::now ());
        sample->sensor_id = 1;
        sample->reading = static_cast<float> (extracted_data & 0xFFFF) * 0.01f;

        // Intrusive O(1) link without dynamic memory allocation
        g_telemetry_queue.link_tail (*sample);
    }
}
return nullptr;
}

// -----
// 5. Telemetry & Audit Thread (Pinned to CPU Core 2)

```

```

// -----
static void*
telemetry_worker_thread_func (void* args)
{
    (void)args;
    trace::printf ("[Core %u] TelemetryWorkerThread started.\n", port_cpu_id ());

    while (true) {
        // Sleep for 100 ms between audit sweeps
        clock_systick::sleep_for (100);

        // D. Scheduler Critical Section (Zero Interrupt Jitter Demonstration)
        // Sets lock_state[Core2] = locked. Preemption on Core 2 is DISABLED,
        // BUT Core 2 hardware interrupts remain ENABLED (zero peripheral latency).
        {
            scheduler::critical_section scs;

            size_t count = 0;
            // Safely iterate through intrusive list without holding hardware spinlock
            for (auto& node : g_telemetry_queue) {
                telemetry_sample_t& sample = static_cast<telemetry_sample_t&> (node);
                // Log telemetry packet...
                count++;
            }

            trace::printf ("[Core %u] Telemetry sweep processed %zu items.\n",
                          port_cpu_id (), count);
        } // Preemption re-enabled on Core 2
    }
    return nullptr;
}

// -----
// 6. Application Initialization & Core Startup (Main Thread on Core 0)
// -----
int
os_main (int argc, char* argv[])
{
    (void)argc;
    (void)argv;

    trace::printf ("\n===== \n");
    trace::printf (" µOS++ IIIE SMP Multi-Core Execution Engine\n");
    trace::printf (" Configured Cores: %u | Active Core: %u\n", OS_NCPU, port_cpu_id ());
    trace::printf ("===== \n");

```

```

// 1. Configure and launch Sensor Processing Thread on CPU Core 1
thread::attributes sensor_attr = thread::initializer;
sensor_attr.th_priority = thread::priority::high;
sensor_attr.th_stack_size_bytes = 2048;
#if defined(OS_USE_SMP_SCHEDULER)
    sensor_attr.th_cpu_affinity = (1U << 1); // Strictly pinned to CPU 1
#endif

    thread sensor_thread{ "sensor-consumer", sensor_processing_thread_func, nullptr, sensor_attr };

// 2. Configure and launch Telemetry Worker Thread on CPU Core 2
thread::attributes telemetry_attr = thread::initializer;
telemetry_attr.th_priority = thread::priority::normal;
telemetry_attr.th_stack_size_bytes = 2048;
#if defined(OS_USE_SMP_SCHEDULER)
    telemetry_attr.th_cpu_affinity = (1U << 2); // Strictly pinned to CPU 2
#endif

    thread telemetry_thread{ "telemetry-audit", telemetry_worker_thread_func, nullptr,
telemetry_attr };

// 3. Main thread continues on CPU Core 0, periodically triggering DMA ISR simulation
for (size_t i = 0; i < 5; ++i) {
    clock_systick::sleep_for (50);
    trace::printf ("[Core %u] Simulating hardware DMA interrupt event #zu...\n",
                    port_cpu_id (), i + 1);
    dma_uart_hardware_isr ();
}

// Allow threads to complete processing
clock_systick::sleep_for (500);

trace::printf ("[Core %u] SMP execution demonstration completed successfully.\n",
                port_cpu_id ());

return 0;
}

```

10.3 State Transition Execution Trace

+-----+ + CHRONOLOGICAL STATE TRANSITION TRACE +-----+				
-----+				
TIME	CORE	EXECUTION CONTEXT	ACTION / PRIMITIVE	MEMORY & HARDWARE
STATE				
+-----+-----+-----+-----+-----+				
+-----+				
T0	C0	os_main()	Launches Threads, sets Affinity	C1 aff=0x02, C2 aff=0x04
T1	C1	SensorWorker	g_data_ready_sem.wait()	C1 enters 5-stage switch to Idle
T2	C2	TelemetryWorker	sleep_for(100)	C2 switches to Idle Core 2
T3	C3	IdleCore[3]	__asm volatile("wfi")	Core 3 enters low-power s tandby
T4	C0	dma_uart_hardware_isr	Enters interrupts::crit_sec	Local CPSID + _smp_klock acquired
T5	C0	dma_uart_hardware_isr	g_data_ready_sem.post()	SensorWorker moved to rea dy list
T6	C0	dma_uart_hardware_isr	port_smp_ipi(CPU_1)	Hardware IPI signal sent to Core 1
T7	C1	IPI Handler on Core 1	Reschedule triggered	Core 1 executes switch_st acks()
T8	C1	SensorWorker (Stage 1)	to->stack_ptr = nullptr	Atomic claim: locked agai nst C2/C3
T9	C1	SensorWorker (Stage 4)	from->stack_ptr = saved_sp	Deferred publish: Idle pu blished
T10	C1	SensorWorker	std::lock_guard<mutex>	Mutex acquired; priority unchanged
T11	C2	TelemetryWorker	scheduler::critical_section	lock_state[2]=1, Core 2 I RQ OPEN
T12	C2	TelemetryWorker	g_telemetry_queue iteration	Traverses node.next() wit hout heap
T13	C2	TelemetryWorker	Exits scheduler::critical_section	lock_state[2]=0, preempti on active
+-----+ -----+				